

Rigorous Mathematical Formulations and Baseline Comparisons for Quantum-Classical Optimization Paradigms

Research Report – Quantum Power Systems Optimization

August 11, 2026

Abstract

This report provides formal mathematical formulations for the synthetic Constrained Quadratic Model (CQM) baseline problem. We detail the mathematical transformations required to map the constrained model onto classical Branch-and-Bound solvers, standard Quadratic Unconstrained Binary Optimization (QUBO) penalty frameworks, parameterized Quantum Approximate Optimization Algorithm (QAOA) Hamiltonians, and the slack-free Powell-Hestenes-Rockafellar Augmented Lagrangian (QPHR-ALM) hybrid framework.

1 Problem Formalization: Synthetic Binary Baseline

The baseline problem is a **Binary Integer Linear Program (BILP)**, which belongs to the NP-hard complexity class. The problem has 6 variables and 3 constraints.

1.1 General Constrained Optimization Model

Formally, we define the optimization problem as minimizing a linear cost function subject to a system of linear inequality constraints:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} = \sum_{j=1}^N c_j x_j \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, \quad \forall i \in \{1, 2, 3\} \\ & \mathbf{x} \in \{0, 1\}^N, \quad N = 6 \end{aligned} \tag{1}$$

where $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6]^T \in \{0, 1\}^6$ is the decision vector, and $\mathbf{c} \in \mathbb{R}^6$ represents objective weights.

1.2 Explicit Algebraic Formulation

Substituting the explicit baseline coefficients yields:

$$\min_{\mathbf{x}} \quad f(\mathbf{x}) = 6x_1 + 3x_2 - 5x_3 - 6x_4 + 4x_5 - 7x_6 \tag{2}$$

subject to the operational inequality constraints:

$$g_1(\mathbf{x}) = -2x_2 - 2x_5 - x_6 + 3 \leq 0 \tag{3}$$

$$g_2(\mathbf{x}) = -x_1 + x_3 - x_4 + 2x_6 \leq 0 \tag{4}$$

$$g_3(\mathbf{x}) = -x_1 + x_3 + x_4 \leq 0 \tag{5}$$

2 Solver Paradigms and Mathematical Transformations

To solve this constrained binary problem across different compute paradigms, the original model must undergo mathematical transformations.

2.1 Exact Classical Solvers

2.1.1 Brute-Force Enumeration

Evaluates the objective energy $f(\mathbf{x})$ over the entire discrete state space $\Omega = \{0, 1\}^N$, filtering out infeasible configurations:

$$\mathbf{x}^* = \arg \min_{\mathbf{x} \in \Omega} \left\{ f(\mathbf{x}) \mid g_i(\mathbf{x}) \leq 0, \forall i \right\}, \quad |\Omega| = 2^6 = 64 \text{ states} \quad (6)$$

2.1.2 Branch-and-Bound (PuLP/CBC)

The PuLP optimization interface delegates the solving of Binary Integer Linear Programs (BILPs) to the COIN-OR Branch-and-Cut (CBC) engine. Because of this, we have no way of reaching the actual solution steps it executed, nevertheless we will try and explain how it works. The solver guarantees exact global optimality by executing a systematic search strategy that couples Linear Programming (LP) continuous relaxations with tree-based spatial partitioning.

A. Continuous LP Relaxation and Bounding Given the original BILP subproblem at node k in the search tree, CBC relaxes the discrete domain $\mathbf{x} \in \{0, 1\}^6$ to a continuous unit hypercube $\mathbf{x} \in [0, 1]^6$. The relaxed linear program LP_k is expressed as:

$$\min_{\mathbf{x}} f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} \quad \text{s.t.} \quad \mathbf{A}\mathbf{x} \leq \mathbf{b}, \quad 0 \leq x_j \leq 1, \forall j \in \{1, \dots, 6\} \quad (7)$$

Because the continuous feasible space strictly contains the discrete feasible set ($\mathcal{F}_{\text{BILP}} \subset \mathcal{F}_{\text{LP}}$), the optimal LP objective energy $z_{\text{LP}}^{(k)}$ establishes a strict mathematical lower bound for node k :

$$z_{\text{LP}}^{(k)} \leq z_{\text{BILP}}^{(k)} \quad (8)$$

B. Tree Partitioning (Branching) If the solution $\mathbf{x}_{\text{LP}}^*$ at node k yields a fractional assignment for a binary variable (i.e., $x_j^* \in (0, 1)$), the subproblem cannot be accepted. CBC selects a fractional branching variable x_j^* according to its node-selection strategy (e.g., strong branching or most-fractional rule) and creates two mutually exclusive child nodes:

$$\text{Left Child: } x_j = 0 \quad \text{and} \quad \text{Right Child: } x_j = 1 \quad (9)$$

C. Pruning Rules (Fathoming) To prevent evaluating all 2^N possible nodes, CBC dynamically prunes (fathoms) active subtrees under three strict conditions:

1. **Pruning by Infeasibility:** The subproblem LP_k has no feasible continuous solution.
2. **Pruning by Optimality:** The solution $\mathbf{x}_{\text{LP}}^*$ satisfies all binary constraints $\mathbf{x}^* \in \{0, 1\}^6$. If $z_{\text{LP}}^{(k)} < z_{\text{incumbent}}$, the global upper bound $z_{\text{incumbent}}$ is updated.
3. **Pruning by Bound:** The node's continuous lower bound exceeds or equals the best-known feasible integer upper bound ($z_{\text{LP}}^{(k)} \geq z_{\text{incumbent}}$), proving that no descendant node can yield a better cost.

2.2 The Bridge from Classical Cost Functions to Quantum Hamiltonians

Physical Quantum Processing Units (QPUs)—whether based on quantum annealing (e.g., D-Wave) or parameterized gate circuits (e.g., IBM Quantum)—cannot directly evaluate arbitrary logical conditionals or constrained inequalities. Instead, quantum hardware operates according to the laws of quantum mechanics, where physical systems naturally evolve toward states that minimize a total energy operator known as the Hamiltonian ($\hat{H}$).

A. Classical Energy Functions to Ising Spin Hamiltonians In quantum computing, an optimization problem is encoded such that the ground state (lowest energy eigenvalue) of a system Hamiltonian corresponds to the global optimal solution $\mathbf{x}^*$. A classical binary variable $x_j \in \{0, 1\}$ is mapped to a physical quantum spin $z_j \in \{-1, +1\}$ via the transformation:

$$x_j = \frac{1 - z_j}{2} \implies z_j = 1 - 2x_j \quad (10)$$

When translated to quantum mechanics, the classical spin values z_j become Pauli-Z operators ($\hat{Z}_j$) acting on the j -th qubit tensor space:

$$\hat{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \hat{Z}|0\rangle = +1|0\rangle, \quad \hat{Z}|1\rangle = -1|1\rangle \quad (11)$$

B. Unconstrained Hamiltonian Formulation Because physical quantum systems evolve unconditionally according to Schrödinger’s equation (or adiabatic transverse-field dissipation), constraints cannot be enforced hard-coded as strict search barriers. Therefore, all inequality constraints $g_i(\mathbf{x}) \leq 0$ must be folded into an unconstrained energy operator:

$$\hat{H}_{\text{total}} = \hat{H}_{\text{objective}} + \hat{H}_{\text{penalty}} \quad (12)$$

C. Taxonomy of Quantum System Hamiltonians Depending on the quantum paradigm utilized, multiple operator structures interact to explore the solution Hilbert space:

1. **Problem / Cost Hamiltonian ($\hat{H}_C$ or $\hat{H}_{\text{problem}}$):** Represents the classical objective function $f(\mathbf{x})$ combined with penalty terms mapped onto diagonal Pauli-Z strings:

$$\hat{H}_C = \sum_{i=1}^N h_i \hat{Z}_i + \sum_{i < j}^N J_{ij} \hat{Z}_i \hat{Z}_j \quad (13)$$

where h_i are single-qubit local longitudinal fields and J_{ij} are two-qubit inter-spin interaction couplings. The ground state eigenvector $|\psi_0\rangle$ of $\hat{H}_C$ directly encodes the optimal binary configuration $\mathbf{x}^*$.

2. **Initial / Driver / Mixer Hamiltonian ($\hat{H}_B$ or $\hat{H}_0$):** An uncoupled transverse-field Hamiltonian that induces quantum fluctuations and enables quantum tunneling across energy barriers. It is constructed using non-diagonal Pauli-X operators:

$$\hat{H}_B = - \sum_{i=1}^N \hat{X}_i, \quad \text{where } \hat{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (14)$$

The unique ground state of $\hat{H}_B$ is an equal superposition state $|+\rangle^{\otimes N} = \frac{1}{\sqrt{2^N}} \sum_{\mathbf{x} \in \{0,1\}^N} |\mathbf{x}\rangle$, representing a uniform probability across all possible 2^N candidate solutions.

3. **Total Time-Dependent Hamiltonian ($\hat{H}(t)$ in Quantum Annealing):** In physical quantum annealing (Adiabatic Quantum Computation), the physical QPU continuously transitions the system energy from the driver Hamiltonian to the problem Hamiltonian over total anneal duration T :

$$\hat{H}(t) = A(t)\hat{H}_B + B(t)\hat{H}_C, \quad t \in [0, T] \quad (15)$$

where $A(0) \gg B(0) \approx 0$ at $t = 0$ (system initialized in equal superposition), and $B(T) \gg A(T) \approx 0$ at $t = T$ (final state collapses into the ground state of $\hat{H}_C$).

2.3 Standard Penalty-Based QUBO Formulations

Physical Quantum Processing Units (QPUs) and gate-based quantum circuits operate on **unconstrained** cost function structures. Standard tools (such as D-Wave's CQM-to-BQM converter) introduce non-negative integer *slack variables* $s_i \in \mathbb{Z}_+$ to map inequality constraints $g_i(\mathbf{x}) \leq 0$ to equality constraints $g_i(\mathbf{x}) + s_i = 0$.

2.3.1 Slack Variable Binarization

Each integer slack variable s_i is bounded by

$$S_i = \max_{\mathbf{x}} g_i(\mathbf{x})$$

and expanded using logarithmic binary encoding:

$$s_i = \sum_{k=0}^{K_i-1} 2^k y_{i,k}, \quad y_{i,k} \in \{0, 1\} \quad (16)$$

where $K_i = \lfloor \log_2(S_i) \rfloor + 1$. For our 3 baseline constraints, this adds 5 auxiliary slack qubits ($y_1 \dots y_5$), expanding the Hilbert space dimension from $N = 6$ to $N_{\text{total}} = 11$.

2.3.2 Standard QUBO Hamiltonian

The unconstrained Quadratic Unconstrained Binary Optimization (QUBO) objective becomes:

$$H_{\text{QUBO}}(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) + \gamma \sum_{i=1}^M \left(g_i(\mathbf{x}) + \sum_{k=0}^{K_i-1} 2^k y_{i,k} \right)^2 \quad (17)$$

where $\gamma > 0$ is a static Lagrange penalty multiplier.

2.4 Quantum Gate-Based QAOA Formulation

Using the Ising mapping $x_j = \frac{1-Z_j}{2}$, where Z_j is the Pauli-Z operator acting on qubit j , the QUBO Hamiltonian maps to an Ising spin operator H_C .

The Quantum Approximate Optimization Algorithm (QAOA) prepares an N -qubit state using alternating layers (p -reps) of cost and mixer unitaries:

$$|\gamma, \beta\rangle = \prod_{l=1}^p e^{-i\beta_l H_B} e^{-i\gamma_l H_C} |+\rangle^{\otimes N} \quad (18)$$

where $H_B = \sum_{j=1}^N X_j$ is the transverse-field mixer operator, and the classical optimizer tunes variational angles (γ, β) to minimize:

$$E(\gamma, \beta) = \langle \gamma, \beta | H_C | \gamma, \beta \rangle \quad (19)$$

2.5 Slack-Free QPHR-ALM Hybrid Formulation

To prevent the exponential qubit explosion caused by classical slack variable expansion ($y_{i,k}$), the framework employs the Powell-Hestenes-Rockafellar (PHR) Augmented Lagrangian Method adapted for quantum optimization (QPHR-ALM).

Instead of introducing K_i auxiliary binary slack variables for each inequality constraint $g_i(\mathbf{x}) \leq 0$, QPHR-ALM incorporates constraints directly into a dynamic penalty-shifted energy function $L(\mathbf{x}; \boldsymbol{\lambda}, \sigma)$:

$$L(\mathbf{x}; \boldsymbol{\lambda}, \sigma) = f(\mathbf{x}) + \frac{1}{2\sigma} \sum_{i=1}^M \left[\max(0, \lambda_i + \sigma g_i(\mathbf{x}))^2 - \lambda_i^2 \right] \quad (20)$$

where $\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_M]^T \geq \mathbf{0}$ is the vector of non-negative Lagrange multipliers, and $\sigma > 0$ represents the penalty stiffness factor.

2.5.1 Iterative Parameter Updates

At outer loop iteration l , the quantum solver (QPU or classical annealer) solves the unconstrained master problem over strictly $N = 6$ qubits:

$$\mathbf{x}^{(l)} = \arg \min_{\mathbf{x} \in \{0,1\}^N} L(\mathbf{x}; \boldsymbol{\lambda}^{(l-1)}, \sigma^{(l-1)}) \quad (21)$$

The classical supervisor then evaluates constraint violations, updating the dual variables λ_i and scaling σ :

$$\lambda_i^{(l)} = \max\left(0, \lambda_i^{(l-1)} + \sigma^{(l-1)} g_i(\mathbf{x}^{(l)})\right), \quad \forall i \in \{1, \dots, M\} \quad (22)$$

$$\sigma^{(l)} = \begin{cases} \eta \sigma^{(l-1)}, & \text{if } R_l \geq \rho R_{l-1} \\ \sigma^{(l-1)}, & \text{otherwise} \end{cases} \quad (23)$$

where $\eta = 1.5$ is the penalty growth factor, $\rho = 0.8$ is the residual reduction threshold, and the overall constraint violation residual R_l is defined according to:

$$R_l = \sqrt{\sum_{i=1}^M \left[\max\left(-\frac{\lambda_i^{(l-1)}}{\sigma^{(l-1)}}, g_i(\mathbf{x}^{(l)})\right) \right]^2} \quad (24)$$

The outer feedback loop terminates when the residual falls below the convergence tolerance $R_l \leq \delta = 10^{-3}$ and the candidate vector $\mathbf{x}^{(l)}$ is strictly feasible ($g_i(\mathbf{x}^{(l)}) \leq 0, \forall i$).

2.5.2 Concrete Numerical Walkthrough (Math Baseline)

To illustrate the dynamic convergence behavior of QPHR-ALM, we trace the exact 3-iteration trajectory observed when executing the synthetic $N = 6$ baseline problem initialized with $\boldsymbol{\lambda}^{(0)} = [0, 0, 0]^T$ and $\sigma_0 = 2.0$:

Iteration 1 ($l = 1$):

- **Parameters:** $\lambda^{(0)} = [0.0, 0.0, 0.0]^T$, $\sigma = 2.0$.
- **BQM Construction:** Unconstrained master BQM $L(\mathbf{x}; \lambda^{(0)}, \sigma_0)$ built on $N = 6$ variables (0 slacks).
- **Quantum Annealer Output:** Sampled state $\mathbf{x}^{(1)} = [1, 0, 0, 1, 0, 1]^T$ (bitstring 100101).
- **Metrics:** Unpenalized objective cost $f(\mathbf{x}^{(1)}) = -7.00$.
- **Feasibility Evaluation:** Constraint $g_1(\mathbf{x}^{(1)}) = -2(0) - 2(0) - 1 + 3 = +2.0 > 0$ (Violated!).
- **Residual & Updates:** $R_1 = 2.0000 > \delta$. Multiplier for g_1 updated to $\lambda_1^{(1)} = \max(0, 0 + 2.0 \times 2.0) = 4.0$.

Iteration 2 ($l = 2$):

- **Parameters:** $\lambda^{(1)} = [4.0, 0.0, 0.0]^T$, $\sigma = 2.0$.
- **BQM Construction:** Penalty terms adjusted; g_1 violation heavily penalized via updated λ_1 .
- **Quantum Annealer Output:** Sampled state shifts to $\mathbf{x}^{(2)} = [1, 1, 0, 1, 0, 1]^T$ (bitstring 110101).
- **Metrics:** Unpenalized objective cost $f(\mathbf{x}^{(2)}) = -4.00$.
- **Feasibility Evaluation:** $g_1 = 0 \leq 0$, $g_2 = 0 \leq 0$, $g_3 = 0 \leq 0$ (Feasible!).
- **Residual & Updates:** $R_2 = 0.0000$. Since $R_2 < \rho R_1$ ($0.0 < 1.6$), σ increased to $\sigma = 1.5 \times 2.0 = 3.0$.

Iteration 3 ($l = 3$ – Final Convergence):

- **Parameters:** $\lambda^{(2)} = [4.0, 0.0, 0.0]^T$, $\sigma = 3.0$.
- **BQM Construction:** Hardened penalty stiffness reinforces feasible state stability.
- **Quantum Annealer Output:** Confirmed optimal assignment $\mathbf{x}^* = [1, 1, 0, 1, 0, 1]^T$ (bitstring 110101).
- **Termination Check:** Residual $R_3 = 0.0000 \leq \delta = 10^{-3}$ and state is fully feasible. The loop terminates successfully in 3 outer iterations.

Table 1: Step-by-Step QPHR-ALM Parameter Evolution Across 3 Outer Iterations

Iter (l)	Bitstring	Cost $f(\mathbf{x})$	Residual R_l	Sigma σ	λ_1	Feasible?
1	100101	-7.00	2.0000	2.0	0.0	No (Violates g_1)
2	110101	-4.00	0.0000	2.0	4.0	Yes
3	110101	-4.00	0.0000	3.0	4.0	Yes (Converged)

Key Mathematical Takeaway: Standard QUBO conversions require 11 physical qubits (6 decision variables +5 auxiliary slack qubits). By adaptively shifting the penalty landscape via λ and σ , QPHR-ALM converged to the global optimum cost $f(\mathbf{x}^*) = -4.00$ in 3 iterations while maintaining a strictly **slack-free footprint of $N = 6$ qubits across every quantum submission.